

Two Extensors of Metrics

by

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Summary. By $\text{Metr}(Z)$ (resp. $\text{Metr}_e(Z)$) we denote the family of all compatible metrics (resp. complete compatible metrics) on a given metrizable space Z . Let A be a closed subset of a metrizable space X . In this note we construct two extensors of metrics $\Phi_1, \Phi_2: \text{Metr}(A) \rightarrow \text{Metr}(X)$ satisfying the conditions

- (i) $\Phi_i(d)|_A = d$ for $d \in \text{Metr}(A)$ and $i=1, 2$.
- (ii) If X is complete-metrizable, then

$$\Phi_i(\text{Metr}_e(A)) \subset \text{Metr}_e(X) \text{ for } i=1, 2.$$

- (iii) Φ_i is a Lipschitz map.
- (iv) If $d_1 \leq d_2$ then $\Phi_2(d_1) \leq \Phi_2(d_2)$ for $d_1, d_2 \in \text{Metr}(A)$.
- (v) If $d_n(x, y) \rightarrow d_0(x, y)$ for $x, y \in A$, then

$$\Phi_2(d_n)(x, y) \rightarrow \Phi_2(d_0)(x, y) \text{ for } x, y \in X.$$

In 1930 Hausdorff [4] proved that if A is a closed subset of a metrizable space X then any compatible metric d on A can be extended to a compatible metric $\tilde{d}$ on X . Since then the problem of extending metrics has been investigated by several authors [2, 8, 6, 5]. Bacon [1] observed that if X is complete-metrizable and d is complete then $\tilde{d}$ can be taken to be complete. The aim of this note is to construct two "canonical" extensors of metrics preserving, e.g. the inequalities of metrics, their point-wise convergence and other properties. The methods used extend those of [7].

Given a metrizable space X . By $\text{Metr}(X)$ (resp. $\text{Metr}_b(X)$, $\text{Metr}_{tb}(X)$, $\text{Metr}_e(X)$, $\text{Pm}(X)$) we denote the set of all compatible metrics (resp. bounded compatible metrics, totally bounded compatible metrics, complete compatible metrics, continuous pseudometrics) on X .

If $d_1, d_2 \in \text{Pm}(X)$ then

$$\|d_1 - d_2\| = \sup \{|d_1(x, y) - d_2(x, y)| : x, y \in X\}$$

defines a metric (which may take the value $+\infty$) on $\text{Pm}(X)$, moreover $\|d\|$ is finite for $d \in \text{Metr}_b(X)$ and $\|d_1 - d_2\|$ may be finite for non-bounded pseudometrics $d_1, d_2 \in \text{Pm}(X)$.

A map $\Phi: \text{Pm}(X) \rightarrow \text{Pm}(Y)$ is Lipschitz iff there exists $C > 0$ such that

$$\|\Phi(d_1) - \Phi(d_2)\| \leq C \|d_1 - d_2\| \text{ for } d_1, d_2 \in \text{Pm}(X).$$

Let X be a metrizable space and A be a closed subset of X . In this note we prove the following two theorems

THEOREM 1. *There exists an extensor $\Phi: \text{Pm}(A) \rightarrow \text{Pm}(X)$ satisfying the conditions*

- (1-i) $\Phi(d)|_A = d$ for $d \in \text{Pm}(A)$.
- (1-ii) $\Phi(\text{Metr}(A)) \subset \text{Metr}(X)$.
- (1-iii) Φ is a Lipschitz map.
- (1-iv) $\Phi(\text{Metr}_b(A)) \subset \text{Metr}_b(X)$.
- (1-v) If X is separable then $\Phi(\text{Metr}_{tb}(A)) = \text{Metr}_{tb}(X)$.
- (1-vi) If X is complete-metrizable then $\Phi(\text{Metr}_c(A)) = \text{Metr}_c(X)$.

THEOREM 2. *There exists an extensor $\Phi: \text{Pm}(A) \rightarrow \text{Pm}(X)$ satisfying the conditions*

- (2-i) $\Phi(d)|_A = d$ for $d \in \text{Pm}(A)$.
- (2-ii) $\Phi(\text{Metr}(A)) \subset \text{Metr}(X)$.
- (2-iii) If $d_1 \leq d_2$ then $\Phi(d_1) \leq \Phi(d_2)$ for $d_1, d_2 \in \text{Pm}(A)$.
- (2-iv) $\Phi(\text{Metr}_b(A)) \subset \text{Metr}_b(X)$.
- (2-v) If X is separable then $\Phi(\text{Metr}_{tb}(A)) \subset \text{Metr}_{tb}(X)$.
- (2-vi) If X is complete-metrizable then $\Phi(\text{Metr}_c(A)) \subset \text{Metr}_c(X)$.
- (2-vii) If $d_n(x, y) \rightarrow d_0(x, y)$ for $x, y \in A$ then

$$\Phi(d_n)(x, y) \rightarrow \Phi(d_0)(x, y) \quad \text{for } x, y \in X.$$

1. Proof of Theorem 1. Consider the locally convex space $E = m(\text{Pm}(A), l^\infty(A))$ of all maps from $\text{Pm}(A)$ into $l^\infty(A)$ equipped with the topology of point-wise convergence. Fix a base point $x_0 \in A$ and define a map $g: A \rightarrow E$ by

$$(1-1) \quad g_x(d)(a) = d(x, a) - d(x_0, a) \quad \text{for } d \in \text{Pm}(A) \text{ and } a \in A.$$

Obviously g is continuous. By Dugundji's theorem [3] there exists a continuous map $\tilde{g}: X \rightarrow E$ such that $\tilde{g}|_A = g$ and $\tilde{g}(X) \subset \text{conv } g(A)$ (the convex hull of $g(A)$ in E).

Fix a compatible metric ρ on X which is complete if X is complete-metrizable and totally bounded if X is separable and define $\Phi_1, \Phi_2, \Phi: \text{Pm}(A) \rightarrow \text{Pm}(X)$ by

$$(1-2) \quad \Phi_1(d)(x, y) = \|g_x(d) - g_y(d)\|$$

$$(1-3) \quad \Phi_2(d)(x, y) = \sup \{ |\theta(d)(x, z) - \theta(d)(x, A) - \theta(d)(y, z) + \theta(d)(y, A)| : z \in X \},$$

where

$$(1-4) \quad \theta(d) = \min \{1, \Phi_1(d) + \rho\}$$

$$(1-5) \quad \Phi(d) = \Phi_1(d) + \Phi_2(d).$$

Obviously Φ satisfies (1-i).

Proof of (1-ii). Since $\theta(\text{Pm}(A)) \subset \text{Metr}(X)$, it suffices to show that $\Phi(d)$ is equivalent to $\theta(d)$ for every $d \in \text{Metr}(A)$.

Suppose that $\theta(d)(x_n, x) \rightarrow 0$. By (1-4) and (1-3) we have $\Phi_i(d)(x_n, x) \rightarrow 0$ for $i = 1, 2$. Thus $\Phi(d)(x_n, x) \rightarrow 0$ by (1-5).

Conversely, assume to the contrary that for some sequence $(x_n) \subset X$ we have $\Phi(d)(x_n, x) \rightarrow 0$ and

$$(1-6) \quad \theta(d)(x_n, x) \geq \delta > 0 \text{ for every } n \in N.$$

Since by (1-3)

$$\theta(d)(x_n, x) \max \{ \theta(d)(x, A), \theta(d)(x_n, A) \} \leq \Phi_2(d)(x_n, x) \leq \Phi(d)(x_n, x) \rightarrow 0$$

we infer that $x \in A$ and $\theta(d)(x_n, A) \rightarrow 0$.

Select a sequence of points $(x'_n) \subset A$ such that

$$(1-7) \quad \theta(d)(x_n, x'_n) \rightarrow 0$$

then $\Phi_1(d)(x_n, x'_n) \rightarrow 0$, by (1-4).

Since $\Phi(d)(x_n, x) \rightarrow 0$, from (1-5) we get $\Phi_1(d)(x_n, x) \rightarrow 0$ and

$$(1-8) \quad d(x'_n, x) = \Phi_1(d)(x'_n, x) \rightarrow 0.$$

Note that $x \in A$, $(x_n) \subset A$ and $d, \theta(d)|_A \in \text{Metr}(A)$, from (1-4) and (1-8) it follows that $\theta(d)(x'_n, x) \rightarrow 0$. Combining this with (1-7) we obtain $\theta(d)(x_n, x) \rightarrow 0$, which contradicts (1-6).

Proof of (1-iii). From (1-1) we get

$$(1-9) \quad \|g_x(d_1) - g_x(d_2)\| \leq 2 \|d_1 - d_2\| \text{ for every } x \in A.$$

Since $\tilde{g}(X) \subset \text{conv } g(A)$, for every $x, y \in X$ we have

$$g_x = \sum_{i=1}^n \lambda_i g_{x_i}, \quad x_i \in A, \quad \lambda_i \geq 0 \text{ for } i=1, \dots, n \text{ and } \sum_{i=1}^n \lambda_i = 1,$$

$$g_y = \sum_{j=1}^m \mu_j g_{y_j}, \quad y_j \in A, \quad \mu_j \geq 0 \text{ for } j=1, \dots, m \text{ and } \sum_{j=1}^m \mu_j = 1.$$

Thus from (1-9) and (1-2) we obtain

$$\|\Phi_1(d_1) - \Phi_1(d_2)\| \leq 4 \|d_1 - d_2\|.$$

On the other hand, from (1-3) and (1-4) we get

$$\|\Phi_2(d_1) - \Phi_2(d_2)\| \leq 4 \|\theta(d_1) - \theta(d_2)\| \leq 4 \|\Phi_1(d_1) - \Phi_1(d_2)\| \leq 16 \|d_1 - d_2\|.$$

Thus by (1-5)

$$\|\Phi(d_1) - \Phi(d_2)\| \leq 20 \|d_1 - d_2\| \text{ for } d_1, d_2 \in \text{Pm}(A)$$

(1-iv) follows from the relation $\tilde{g}(X) \subset \text{conv } g(A)$.

Proof of (1-v). Assume that ρ and d are totally bounded. Since

$$\|g_x(d) - g_y(d)\| = d(x, y)$$

the set

$$g(A)d = \{g(d) : d \in A\} \subset l^\infty(A)$$

is totally bounded. Thus $\tilde{g}(X) d \subset \text{conv } g(A) d$ is also totally bounded. Hence by (1-2) $\Phi_1(d)$ is totally bounded. Consequently by (1-4) $\theta(d)$ is totally bounded. Since $\Phi_2(d) \leq 2\theta(d)$ we infer that

$$\Phi(d) = \Phi_1(d) + \Phi_2(d)$$

is totally bounded.

Proof of (1-vi). By (1-4) if ρ is complete then $\theta(d)$ is also a complete metric for every $d \in \text{Pm}(A)$. Thus from the equivalence of $\Phi(d)$ and $\theta(d)$ it suffices to show that if $\Phi(d)(x_n, x_m) \rightarrow 0$ then $\theta(d)(x_n, x_m) \rightarrow 0$ for every $d \in \text{Metr}_c(A)$.

If it is not the case then there exist two subsequences $(a_n) \subset (x_n)$ and $(b_n) \subset (x_n)$ such that

$$(1-10) \quad \theta(d)(a_n, b_n) \geq \delta > 0 \text{ for every } n \in N.$$

Since by (1-3)

$$\theta(d)(a_n, b_n) \theta(d)(a_n, A) \leq \Phi_2(d)(a_n, b_n) \leq \Phi(d)(a_n, b_n) \rightarrow 0$$

we infer that there exists a sequence (y_n) of points in A such that

$$(1-11) \quad \theta(d)(a_n, y_n) \rightarrow 0.$$

Thus from (1-4) $\Phi_1(d)(a_n, y_n) \rightarrow 0$. Since $\Phi_1(d)|_{A=d}$ is complete and since

$$\begin{aligned} \Phi_1(d)(y_n, y_m) &\leq \Phi_1(d)(y_n, a_n) + \Phi_1(d)(a_n, a_m) + \Phi_1(d)(a_m, y_m) \leq \\ &\leq \Phi_1(d)(y_n, a_n) + \Phi(d)(a_n, a_m) + \Phi_1(d)(a_m, y_m) \rightarrow 0 \end{aligned}$$

we infer that there exists a point $x_0 \in A$ such that

$$\Phi(d)(y_n, x_0) = \Phi_1(d)(y_n, x_0) = d(y_n, x_0) \rightarrow 0.$$

Thus $\theta(d)(y_n, x_0) \rightarrow 0$. Hence from (1-11) we get $\theta(d)(a_n, x_0) \rightarrow 0$. Consequently $\Phi(d)(a_n, x_0) \rightarrow 0$. Since (x_n) is a $\Phi(d)$ -Cauchy sequence and $(a_n) \subset (x_n)$ we infer that $\Phi(d)(x_n, x_0) \rightarrow 0$. Thus $\theta(d)(x_n, x_0) \rightarrow 0$. This contradicts (1-10) and concludes the proof of Theorem 1.

2. Proof of Theorem 2. For every $d \in \text{Pm}(A)$ let $\text{Lip}(A, d)$ denote the set of all Lipschitz maps $f: (A, d) \rightarrow \mathbb{R}^1$ vanishing at $a_0 \in A$ equipped with the norm

$$\|f\|_d = \inf \{C: f \text{ is a } C\text{-Lipschitz map}\}.$$

Let $L(A)$ denote the linear space spanned formally by A . For every $d \in \text{Pm}(A)$ and $x = \sum_{i=1}^n \lambda_i a_i \in L(A)$, put

$$\|x\|_d = \sup \left\{ \left| \sum_{i=1}^n \lambda_i f(a_i) \right| : f \in \text{Lip}(A, d), \|f\|_d \leq 1 \right\}.$$

Then $L(A)$ becomes a locally convex space. By Dugundji's theorem [3] there exists a continuous map $\varphi: X \rightarrow L(A)$ such that $\varphi(a) = a$ for every $a \in A$ and $\varphi(X) \subset \text{conv}(A) \subset L(A)$.

Fix a compatible metric ρ on X which is complete if X is complete-metrizable and totally bounded if X is separable and define $\Phi_1, \Phi_2, \Phi: \text{Pm}(A) \rightarrow \text{Pm}(X)$ by

$$(2-1) \quad \Phi_1(d)(x, y) = \|\varphi(x) - \varphi(y)\|_d$$

$$(2-2) \quad \Phi_2(d)(x, y) = \sup \{ \|f(x) - f(y)\| : f \in \text{Lip}(X, e), f|_A = 0 \text{ and } \|f\|_e \leq 1 \},$$

where $e = \min \{1, \rho + \Phi_1(d)\}$

$$(2-3) \quad \Phi(d) = \Phi_1(d) + \Phi_2(d).$$

It is easy to see that Φ satisfies conditions (2-i) and (2-iii). The proofs of (2-ii), (2-iv), (2-v) and (2-vi) are similar to the proofs of (1-ii), (1-iv), (1-v) and (1-vi), respectively. Let us check (2-vii).

Assume that $d_n(x, y) \rightarrow d_0(x, y)$ for $x, y \in A$. We have to show that $\Phi_i(d_n)(x, y) \rightarrow \Phi_i(d_0)(x, y)$ for $x, y \in X$ and $i = 1, 2$.

Take $x_0, y_0 \in X$. Then we have

$$\varphi(x_0) - \varphi(y_0) = \sum_{i=1}^k \lambda_i a_i,$$

where $a_1, a_2, \dots, a_k \in A$.

The case $i = 1$. For every $\varepsilon > 0$, take $p > \Phi_1(d_0)(x_0, y_0)$ and let $n_0 \in N$ be such that

$$(2-4) \quad \left(1 - \frac{\varepsilon}{p}\right) d_0(a, b) \leq d_n(a, b) \leq \left(1 + \frac{\varepsilon}{p}\right) d_0(a, b)$$

for every $n \geq n_0$ and $a, b \in A_0 = \{a_1, \dots, a_k\}$.

From (2-4) we get for $n \geq n_0$

$$\begin{aligned} \Phi_1(d_n)(x_0, y_0) &= \sup \left\{ \left\| \sum_{i=1}^k \lambda_i f(a_i) \right\| : f \in \text{Lip}(A, d_n), \|f\|_{d_n} \leq 1 \right\} = \\ &= \sup \left\{ \left\| \sum_{i=1}^k \lambda_i f(a_i) \right\| : f \in \text{Lip}(A_0, d_n), \|f\|_{d_n} \leq 1 \right\} \leq \\ &\leq \sup \left\{ \left\| \sum_{i=1}^k \lambda_i f(a_i) \right\| : f \in \text{Lip}(A_0, d_0), \|f\|_{d_0} \leq 1 + \frac{\varepsilon}{p} \right\} = \\ &= \left(1 + \frac{\varepsilon}{p}\right) \sup \left\{ \left\| \sum_{i=1}^k \lambda_i f(a_i) \right\| : f \in \text{Lip}(A_0, d_0), \|f\|_{d_0} \leq 1 \right\} = \\ &= \left(1 + \frac{\varepsilon}{p}\right) \sup \left\{ \left\| \sum_{i=1}^k \lambda_i f(a_i) \right\| : f \in \text{Lip}(A, d_0), \|f\|_{d_0} \leq 1 \right\} = \\ &= \left(1 + \frac{\varepsilon}{p}\right) \Phi_1(d_0)(x_0, y_0) < \Phi_1(d_0)(x_0, y_0) + \varepsilon. \end{aligned}$$

Similarly,

$$\Phi_1(d_n)(x_0, y_0) > \Phi_1(d_0)(x_0, y_0) - \varepsilon \text{ for every } n \geq n_0.$$

The case $i=2$. Let $e_n(x, y) = \rho(x, y) + \Phi_1(d_n)(x, y)$. By the case $i=1$ we get

$$e_n(x, y) \rightarrow e_0(x, y) \text{ for every } x, y \in X.$$

For every $\varepsilon > 0$, take an $n_0 \in \mathbb{N}$ such that

$$(2-5) \quad \left(1 - \frac{\varepsilon}{p}\right) e_0(x, y) \leq e_n(x, y) \leq \left(1 + \frac{\varepsilon}{p}\right) e_0(x_0, y_0) \text{ for every } n \geq n_0,$$

where $p > \Phi_2(d_0)(x_0, y_0)$.

Thus for every $n \geq n_0$ we have

$$\begin{aligned} \Phi_2(d_n)(x_0, y_0) &= \sup \{ |f(x_0) - f(y_0)| : f \in \text{Lip}(X, e_n), f|_A = 0 \text{ and } \|f\|_{e_n} \leq 1 \} \leq \\ &\leq \sup \left\{ |f(x_0) - f(y_0)| : f \in \text{Lip}(X, e_0), f|_A = 0 \text{ and } \|f\|_{e_0} \leq 1 + \frac{\varepsilon}{p} \right\} = \\ &= \left(1 + \frac{\varepsilon}{p}\right) \sup \{ |f(x_0) - f(y_0)| : f \in \text{Lip}(X, e_0), f|_A = 0 \text{ and } \|f\|_{e_0} \leq 1 \} = \\ &= \left(1 + \frac{\varepsilon}{p}\right) \Phi_2(d_0)(x_0, y_0) < \Phi_2(d_0)(x_0, y_0) + \varepsilon. \end{aligned}$$

Similarly,

$$\Phi_2(d_n)(x_0, y_0) > \Phi_2(d_0)(x_0, y_0) - \varepsilon \text{ for every } n \geq n_0.$$

This completes the proof of Theorem 2.

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Нгуен Ван Khue, Нгуен То Ну, О двух экстенсорах метрик.

Через $\text{Metr}(Z)$ (соответственно, $\text{Metr}_c(Z)$) мы обозначаем семейство всех компактизуемых метрик (соответственно, вполне компактизуемых метрик) на данном метризуемом пространстве Z . Пусть A замкнутое подмножество метризуемого пространства X . В данной работе строятся два экстенсора метрик $\Phi_1, \Phi : \text{Metr}(A) \rightarrow \text{Metr}(X)$, удовлетворяющих условиям:

- (i) $\Phi_i(d)|_A = d$, где $d \in \text{Metr}(A)$, $i=1, 2$;
- (ii) если X вполне метризуемо, то $\Phi_i(\text{Metr}_c(A)) \subset \text{Metr}_c(X)$ $i=1, 2$;
- (iii) Φ_1 есть Липшицово отображение;
- (iv) если $d_1 \leq d$, то $\Phi(d_1) \leq \Phi(d)$ где $d_1, d \in \text{Metr}(A)$;
- (v) если $d_n(x, y)$, где $x, y \in A$, то $\Phi(d_n)(x, y) \leq \Phi(d)(x, y)$, где $x, y \in X$.